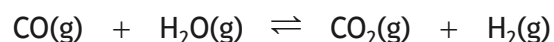


8. Fizzy drinks are made by adding carbon dioxide gas, preservative, colouring and flavouring to water.

(a) Carbon dioxide for fizzy drinks can be produced using the water-gas shift reaction.



(i) A catalyst for this reaction is copper(II) oxide.

Complete the table by circling one option on each line to show the effect of copper(II) oxide on the reaction.

2

(An additional table can be found on *page 33*.)

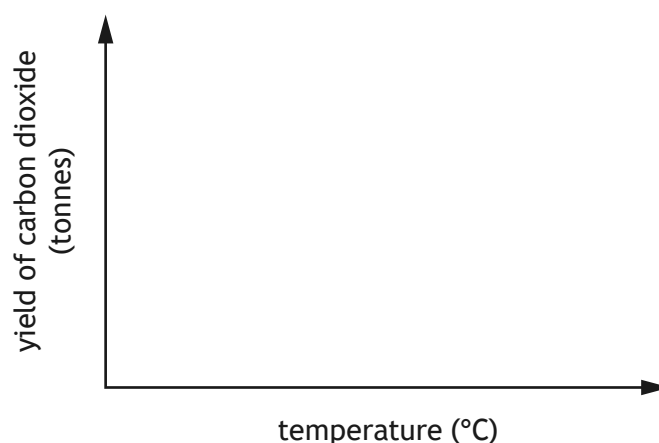
Feature of reaction	Effect of catalyst
Rate of forward reaction	increase/decrease/no effect
Rate of reverse reaction	increase/decrease/no effect
Position of equilibrium	moves to right/moves to left/no effect

(ii) The water-gas shift reaction is exothermic.

Draw a line on the axes below to show how the yield of carbon dioxide would vary with increasing temperature.

1

(An additional diagram can be found on *page 33*.)



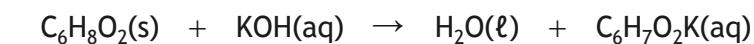
[Turn over



8. (continued)

- (b) A preservative added to some fizzy drinks is made by reacting sorbic acid and potassium hydroxide.

In an experiment, 7 g of sorbic acid, $\text{C}_6\text{H}_8\text{O}_2$, is reacted with 250 cm^3 of potassium hydroxide solution, concentration 0.5 mol l^{-1} .



$GFM = 112 \text{ g}$

Show, by calculation, that sorbic acid is the limiting reactant.

2



* X 8 1 3 7 6 0 1 2 2 *

8. (continued)

- (c) Ammonium ferric citrate ($GFM = 261.8$ g) gives some drinks a characteristic orange colour. A typical drink contains 0.002% of ammonium ferric citrate.

A 1% solution contains 1 g made up to 100 cm^3 of solution.

Calculate the number of moles of ammonium ferric citrate required to make 330 cm^3 of this fizzy drink.

2

[Turn over

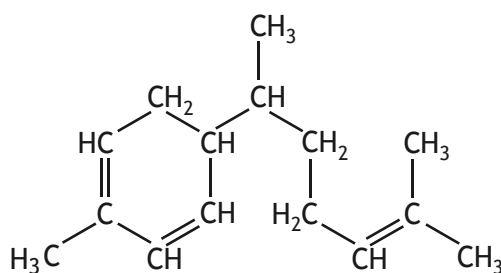


8. (continued)

(d) Ginger root is used as a flavouring for some fizzy drinks.

(i) Ginger oil is an essential oil obtained from ginger root.

Zingiberene is one of the main components in this essential oil.



zingiberene

(A) State one property of an essential oil.

1

(B) Zingiberene is formed from isoprene units.

(I) Name the type of compound formed when isoprene units join together.

1

(II) Isoprene is also called 2-methyl-1,3-butadiene.

Draw a structural formula for isoprene.

1

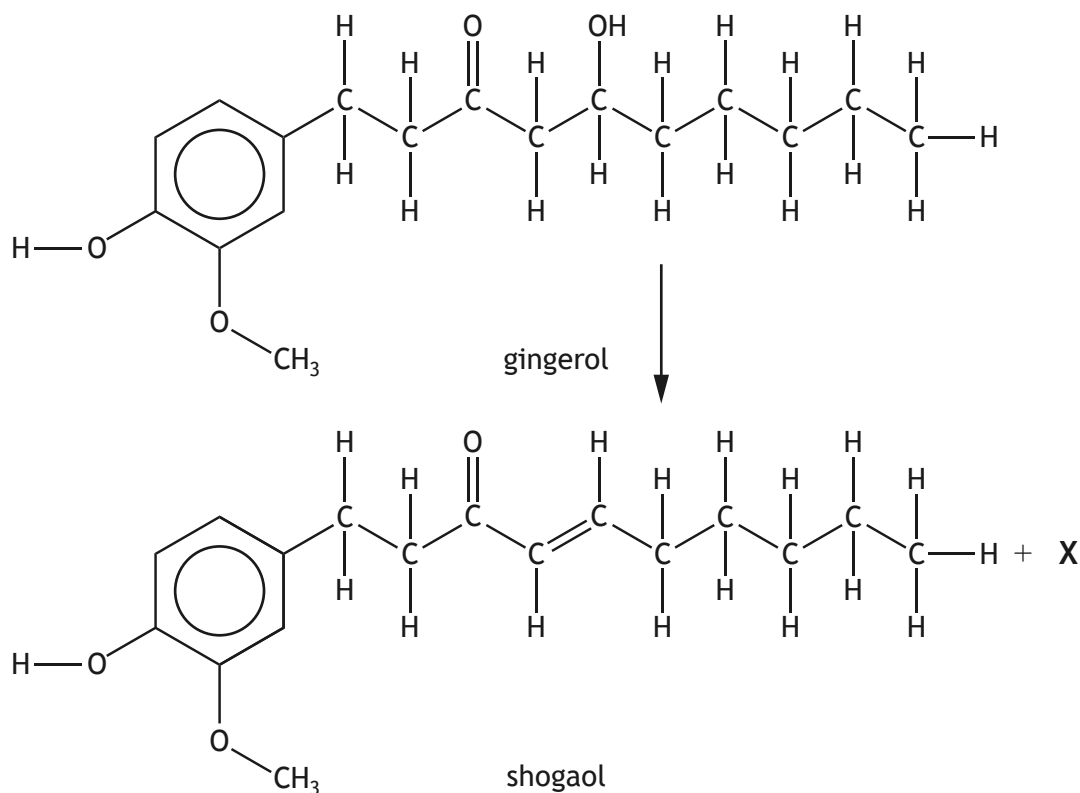
(III) State the number of isoprene units in a zingiberene molecule.

1



8. (d) (continued)

- (ii) Gingerol is another compound found in ginger root. Gingerol can form the compound shogaol as shown.



(A) Name product X.

1

(B) Name two functional groups present in gingerol and shogaol that are not present in zingiberene.

1

[Turn over



* X 8 1 3 7 6 0 1 2 5 *